

05-examples

2520090128

1. INNER JOIN

The screenshot shows a database IDE with the following components:

- SQL Editor:** Contains the following SQL code:

```
29 (2,'Surgery',7),
30 (3,'Psychiatry',9);
31
32 • SELECT d.name AS Department,
33        p.name AS Head_Physician
34 FROM department d
35 JOIN physician p
36 ON d.head = p.employeeid;
```
- Result Grid:** Displays the query results in a table:

Department	Head_Physician
General Medicine	Pratik Kumar
Surgery	Arnav Chitkena
Psychiatry	Molly Clock
- Action Output:** Shows the execution log with the following entries:

#	Time	Action	Message
1	22:03:05	USE bookflow_db	0 row(s) affected
2	22:03:08	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR...	0 row(s) affected
3	22:04:11	INSERT INTO physician VALUES (1,'Mohammad Rehan','Staff Internist','111111111'), (2,'Akhl Karuturi','Attendin...	9 row(s) affected Records: 1
4	22:04:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
5	22:04:16	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records: 1
6	22:09:25	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d....	3 row(s) returned

2. WHERE + LIKE

```

33      p.name AS Head_Physician
34  FROM department d
35  JOIN physician p
36  ON d.head = p.employeeid;
37
38  • SELECT *
39  FROM physician
40  WHERE position = 'Surgical Attending Physician';
41

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
▶	3	Rohit Upputuri	Surgical Attending Physician	333333333
	6	Sreekar Varanasi	Surgical Attending Physician	666666666
	7	Arnav Chitkena	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

physician 2 x

#	Time	Action	Message
✓	2 22:03:08	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR(100))	0 row(s) affected
✓	3 22:04:11	INSERT INTO physician VALUES (1,'Mohammad Rehan','Staff Internist','1111111111'), (2,'Akhl Karuturi','Atten...)	9 row(s) affected Recon
✓	4 22:04:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓	5 22:04:16	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Recon
✓	6 22:09:25	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	7 22:26:40	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples x

```

37
38  • SELECT *
39  FROM physician
40  WHERE position = 'Surgical Attending Physician';
41
42  • SELECT *
43  FROM physician
44  WHERE name LIKE 'John%';

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
*	NULL	NULL	NULL	NULL

physician 3 x

#	Time	Action	Message
✓	3 22:04:11	INSERT INTO physician VALUES (1,'Mohammad Rehan','Staff Internist','1111111111'), (2,'Akhl Karuturi','Atten...)	9 row(s) affected Records:
✓	4 22:04:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓	5 22:04:16	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records:
✓	6 22:09:25	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	7 22:26:40	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓	8 22:27:08	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned

3. COUNT + COUNT(DISTINCT)

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples* x

Limit to 1000 rows

```

40 WHERE position = 'Surgical Attending Physician';
41
42 • SELECT *
43 FROM physician
44 WHERE name LIKE 'John%';
45
46 • SELECT COUNT(*) AS Total_Physicians
47 FROM physician;
48

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Total_Physicians
9

Result 4 x

Output

Action Output

#	Time	Action	Message
4	22:04:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
5	22:04:16	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected Records
6	22:09:25	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
7	22:26:40	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
8	22:27:08	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
9	22:28:37	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples* x

Limit to 1000 rows

```

43 FROM physician
44 WHERE name LIKE 'John%';
45
46 • SELECT COUNT(*) AS Total_Physicians
47 FROM physician;
48
49 • SELECT COUNT(DISTINCT position) AS Different_Positions
50 FROM physician;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Different_Positions
7

Result 5 x

Output

Action Output

#	Time	Action	Message
5	22:04:16	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)	3 row(s) affected
6	22:09:25	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
7	22:26:40	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
8	22:27:08	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
9	22:28:37	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
10	22:28:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned

4. GROUP BY + HAVING

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples*

```

48
49 • SELECT COUNT(DISTINCT position) AS Different_Positions
50 FROM physician;
51
52 • SELECT position,
53       COUNT(*) AS Total_Employees
54 FROM physician
55 GROUP BY position;

```

Result Grid Filter Rows: Export: Wrap Cell Content:

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 6 x

Output

Action Output

#	Time	Action	Message
6	22:09:25	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
7	22:26:40	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
8	22:27:08	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
9	22:28:37	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
10	22:28:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
11	22:30:28	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
57		SELECT position,	
58		COUNT(*) AS Total_Employees	
59		FROM physician	
60		GROUP BY position	
61		HAVING COUNT(*) > 1;	

Result Grid Filter Rows: Export: Wrap Cell Content:

position	Total_Employees
Surgical Attending Physician	3

Result 7 x

Output

Action Output

#	Time	Action	Message
7	22:26:40	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
8	22:27:08	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
9	22:28:37	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
10	22:28:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
11	22:30:28	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
12	22:30:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

5. ORDER BY

```

63 • SELECT *
64 FROM physician
65 ORDER BY name ASC;
66

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
▶	2	Akhil Karuturi	Attending Physician	222222222
	7	Arnav Chitkena	Surgical Attending Physician	777777777
	8	Keith Dudemeister	MD Resident	888888888
	1	Mohammad Rehan	Staff Internist	111111111
	9	Molly Clock	Attending Psychiatrist	999999999
	4	Pratik Kumar	Senior Attending Physician	444444444
	3	Rohit Upputuri	Surgical Attending Physician	333333333
	6	Sreekar Varanasi	Surgical Attending Physician	666666666
	5	Yugandar Peddi	Head Chief of Medicine	555555555
*	NULL	NULL	NULL	NULL

physician 8 x

Output

Action Output

#	Time	Action	Message
✓ 8	22:27:08	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned
✓ 9	22:28:37	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 10	22:28:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	22:30:28	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:30:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:31:34	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

```

67 • SELECT *
68 FROM physician
69 ORDER BY employeeid DESC;
70

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
▶	9	Molly Clock	Attending Psychiatrist	999999999
	8	Keith Dudemeister	MD Resident	888888888
	7	Arnav Chitkena	Surgical Attending Physician	777777777
	6	Sreekar Varanasi	Surgical Attending Physician	666666666
	5	Yugandar Peddi	Head Chief of Medicine	555555555
	4	Pratik Kumar	Senior Attending Physician	444444444
	3	Rohit Upputuri	Surgical Attending Physician	333333333
	2	Akhil Karuturi	Attending Physician	222222222
	1	Mohammad Rehan	Staff Internist	111111111
*	NULL	NULL	NULL	NULL

physician 9 x

Output

Action Output

#	Time	Action	Message
✓ 9	22:28:37	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 10	22:28:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	22:30:28	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:30:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:31:34	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:32:04	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned

6. SUBQUERIES — IN + NOT IN

```

71 • SELECT *
72 FROM physician
73 WHERE employeeid IN
74 (
75     SELECT head
76     FROM department
77 );
78

```

Result Grid

	employeeid	name	position	ssn
▶	4	Pratik Kumar	Senior Attending Physician	444444444
	7	Arnav Chitkena	Surgical Attending Physician	777777777
	9	Molly Clock	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 10 ×

Output

Action Output

#	Time	Action	Message
✓ 10	22:28:48	SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 11	22:30:28	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:30:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:31:34	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:32:04	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:33:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned

```

79 • SELECT *
80 FROM physician
81 WHERE employeeid NOT IN
82 (
83     SELECT head
84     FROM department
85 );
86

```

Result Grid

	employeeid	name	position	ssn
▶	1	Mohammad Rehan	Staff Internist	111111111
	2	Akhil Karuturi	Attending Physician	222222222
	3	Rohit Upputuri	Surgical Attending Physician	333333333
	5	Yugandar Peddi	Head Chief of Medicine	555555555
	6	Sreekar Varanasi	Surgical Attending Physician	666666666
	8	Keith Dudemeister	MD Resident	888888888
*	NULL	NULL	NULL	NULL

physician 11 ×

Output

Action Output

#	Time	Action	Message
✓ 11	22:30:28	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000	7 row(s) returned
✓ 12	22:30:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:31:34	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:32:04	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:33:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:33:22	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned

7. GROUP BY + ORDER BY + LIKE

```

87 • SELECT position,
88       COUNT(*) AS Total_Employees
89 FROM physician
90 GROUP BY position
91 ORDER BY Total_Employees DESC;
92

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

Result 12 x

Output

Action Output

#	Time	Action	Message
✓ 12	22:30:50	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓ 13	22:31:34	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:32:04	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:33:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:33:22	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:34:00	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned

```

93 • SELECT *
94 FROM physician
95 WHERE position LIKE '%Attending%';
96

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
▶	2	Akhil Karuturi	Attending Physician	222222222
	3	Rohit Upputuri	Surgical Attending Physician	333333333
	4	Pratik Kumar	Senior Attending Physician	444444444
	6	Sreekar Varanasi	Surgical Attending Physician	666666666
	7	Arnav Chitkena	Surgical Attending Physician	777777777
	9	Molly Clock	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

physician 13 x

Output

Action Output

#	Time	Action	Message
✓ 13	22:31:34	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓ 14	22:32:04	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:33:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:33:22	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:34:00	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:34:16	SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000	6 row(s) returned

8. JOIN + WHERE / JOIN + ORDER BY

```

97 • SELECT p.name
98 FROM physician p
99 JOIN department d
100 ON p.employeeid = d.head
101 WHERE d.name = 'Surgery';
102

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

name
Arnav Chitkena

Result 14 x

Output

Action Output

#	Time	Action	Message
✓ 14	22:32:04	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned
✓ 15	22:33:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:33:22	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:34:00	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:34:16	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
✓ 19	22:34:51	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned

```

103 • SELECT d.name,
104         p.name
105 FROM department d
106 JOIN physician p
107 ON d.head = p.employeeid
108 ORDER BY d.name;
109

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

name	name
General Medicine	Pratik Kumar
Psychiatry	Molly Clock
Surgery	Arnav Chitkena

Result 15 x

Output

Action Output

#	Time	Action	Message
✓ 15	22:33:05	SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000	3 row(s) returned
✓ 16	22:33:22	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:34:00	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:34:16	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
✓ 19	22:34:51	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 20	22:35:11	SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...	3 row(s) returned

9. NOT LIKE + COUNT + HAVING

```

110 • SELECT *
111 FROM physician
112 WHERE position NOT LIKE '%Surgical%';
113

```

Result Grid				
Filter Rows:				
Edit:				
Export/Import:				
Wrap Cell Content:				
	employeeid	name	position	ssn
▶	1	Mohammad Rehan	Staff Internist	111111111
	2	Akhil Karuturi	Attending Physician	222222222
	4	Pratik Kumar	Senior Attending Physician	444444444
	5	Yugandar Peddi	Head Chief of Medicine	555555555
	8	Keith Dudemeister	MD Resident	888888888
	9	Molly Clock	Attending Psychiatrist	999999999
•	NULL	NULL	NULL	NULL

physician 16 x

Output

Action Output

#	Time	Action	Message
✓ 16	22:33:22	SELECT * FROM physician WHERE employeeid NOT IN (SELECT head FROM department) LIMIT 0, 1...	6 row(s) returned
✓ 17	22:34:00	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned
✓ 18	22:34:16	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
✓ 19	22:34:51	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 20	22:35:11	SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...	3 row(s) returned
✓ 21	22:35:40	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
113 •		SELECT position,	
114		COUNT(*) AS Total	
115		FROM physician	
116		GROUP BY position	
117		HAVING COUNT(*) >= 2;	
118			

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	position	Total
▶	Surgical Attending Physician	3

Result 17 x

Output				
Action Output				
#	Time	Action	Message	
✓ 17	22:34:00	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...	7 row(s) returned	
✓ 18	22:34:16	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned	
✓ 19	22:34:51	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned	
✓ 20	22:35:11	SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...	3 row(s) returned	
✓ 21	22:35:40	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned	
✓ 22	22:35:57	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned	

10.EXISTS + MAX + MIN

```

119 • SELECT *
120 FROM department d
121 WHERE EXISTS
122 (
123     SELECT 1
124     FROM physician p
125     WHERE p.employeeid = d.head
126 );
127

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
*	NULL	NULL	NULL

department 18 x

Output

Action Output

#	Time	Action	Message
✓ 18	22:34:16	SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000	6 row(s) returned
✓ 19	22:34:51	SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery...	1 row(s) returned
✓ 20	22:35:11	SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...	3 row(s) returned
✓ 21	22:35:40	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓ 22	22:35:57	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
✓ 23	22:36:21	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned

```

128 • SELECT MAX(employeeid) AS Highest_Employee_ID
129 FROM physician;
130

```

Result Grid Filter Rows: Export: Wrap Cell Content:

Highest_Employee_ID
▶ 9

Result 20 x

Output

Action Output

#	Time	Action	Message
✓ 20	22:35:11	SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...	3 row(s) returned
✓ 21	22:35:40	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓ 22	22:35:57	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...	1 row(s) returned
✓ 23	22:36:21	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeei...	3 row(s) returned
✓ 24	22:36:44	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
✓ 25	22:38:26	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned

```

131 • SELECT MIN(employeeid) AS Lowest_Employee_ID
132 FROM physician;
133

```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

Lowest_Employee_ID
1

Result 21 x

Output

 Action Output 

	#	Time	Action	Message
✓	21	22:35:40	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000	6 row(s) returned
✓	22	22:35:57	SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMIT 0, 1000	1 row(s) returned
✓	23	22:36:21	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid = d.department_id)	3 row(s) returned
✓	24	22:36:44	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
✓	25	22:38:26	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
✓	26	22:38:36	SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned